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1.1 Introduction

Planning of the power distribution is one of the most important components of power system. Sub transmission is the efficient and economical method of power between power distribution and transmission. The power is transmitted through the overhead line distribution system and underground cable. These types of transmission have their own electrical properties and have significant effects during power distribution.

In this chapter, the properties of various types of electrical power stages are specifically explained as high voltage and low voltage transmission, overhead transmission and underground cables. From this lesson, various types of poles, insulators, and protective devices are explained clearly.

1.2 Power transmission

Electrical Power transmission is one of the major concerns in the power supply system. There are three main areas of power system. They are power generation, transmission and distribution. Transmission is done by

i. Overhead lines
ii. Underground cables

1.2.1 Power generation station

Electricity is produced in power plants, which goes through different levels to reach consumers. Generated power is stepped up and stepped down by using transformers and it is transmitted. In Tamil Nadu, power plants are far from consumers who use power. For example, the important places of power plants are located at Kalpakkam, Koodankulam (Nuclear power station), Neyveli (Thermal power station), Bhavani sagar Dam, Pykara, Kunda Dam (Water power station) and Kamuti (Solar power station) etc, The generated power is distributed to the consumers through the substations.

1.2.2 Various system of power transmission

i. DC system
   a) DC two wire
   b) DC two wire with mid point earthed
   c) DC three wire system
ii. Single phase AC system
   a) Single phase two wire
   b) Single phase two wire with midpoint earthed
   c) Single phase three wire

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